

Fm16
Leon MacQueen Dist 3

Consolidation

$$1) X \sim N(\mu, 3^2)$$

$$P(X > 16) = 0.15$$

$$P(Z > \frac{16 - \mu}{3}) = 0.15$$

$$P(Z < \frac{16 - \mu}{3}) = 0.85$$

$$\frac{16 - \mu}{3} = 1.0364$$

$$16 - \mu = 3.109$$

$$\mu = 12.891 \text{ cm}$$

$$P(X > 12) = \underline{0.617}$$

$$2) X \sim N(\mu, \sigma^2)$$

$$P(N > 428) = 0.18$$

$$P(N < 428) = 0.82$$

$$P(N > 416) = 0.30$$

$$P(N < 416) = 0.70$$

$$P(Z < \frac{428 - \mu}{\sigma}) = .82$$

$$P(Z < \frac{416 - \mu}{\sigma}) = .70$$

$$428 - \mu = 0.91536 \sigma$$

$$416 - \mu = 0.52440 \sigma$$

$$\sigma = 30.69$$

$$\mu = 399.9$$

$$3 \quad X \sim N(\mu, \sigma)$$

$$P(X > 35) = 0.75$$

$$Z < \frac{35 - \mu}{\sigma} = .75$$

$$P(X > 53) = .25$$

$$Z < \frac{53 - \mu}{\sigma} = .15$$

$$\frac{35 - \mu}{\sigma} = -0.674$$

$$\frac{53 - \mu}{\sigma} = 0.674$$

$$\sigma = 13.35 \rightarrow \sigma^2 = 178$$

$$\mu = 44$$

4 500 = nominal weight

X = weight of packet

$$X \sim N(508.3, 4.8^2)$$

$$P(X < 500) = 0.0418$$

Y = new weight

$$Y \sim N(\mu, 4.8^2)$$

$$P(Y < 500) = 0.01 \times \frac{1}{2} = 0.005$$

$$P\left(Z < \frac{500 - \mu}{4.8}\right) = 0.005$$

$$\frac{500 - \mu}{4.8} = -2.576$$

$$500 - \mu = -12.36$$

$$\mu = 512.4$$

$$5) X = \text{length of rod}$$

$$= N(\mu, \sigma)$$

$$P(X \geq 30.6) = 0.30$$

$$P(X \leq 24.2) = 0.15$$

$$P\left(Z \leq \frac{30.6 - \mu}{\sigma}\right) = 0.70$$

$$P\left(Z \leq \frac{24.2 - \mu}{\sigma}\right) = 0.15$$

$$0.5244\sigma + \mu = 30.6 \quad \sigma = 0.848$$

$$-1.036\sigma + \mu = 24.2 \quad \mu = 30.1$$

$$ii) P(x > 24.9) = 0.588$$

$$iii) P(24.8 \leq x \leq 30.5) = 0.3028$$

$$1 - 0.3028 = 0.6972$$

69.7%

Exam Q

a) 20 measure = 300 days
 1 msq = 15 days 1 li square = 0.6 days

$\therefore$ 60 days when income between 4000 and 6000

b) $3 \times 15 + 8 \times 25 + 1 \times 15 = 185$ small squares
 < 111 days

$$c) \frac{1800 \times 30 + 2500 \times 40 + 3250 \times 75 + 3750 \times 45 + 5000 \times 60}{360} = 3275$$

d) $x \sim N(3275, 1060^2)$

$$P(4000 < x < 6000) = 0.2414$$

$$300 \times 0.2414 = 72.57$$

e) $3275 + 2 \times 1060 = 5395$
 $3275 - 2 \times 1060 = 1155$

$$P(x > 5395) = 0.023$$

$$P(x < 1155) = 0.027$$

$\therefore$ total probability is $0.05 = 5\%$ - quite high
 $\therefore$ not a good fit

$$Z \quad x \sim N(56, \sigma^2)$$

$$P(x > 53) = 0.7$$

$$P(z < \frac{53 - 56}{\sigma}) = 0.3$$

$$-3 = -0.524 \sigma$$

$$\sigma = 5.7208$$

- a) 30%
 b) 5.78%

$$\begin{aligned} \textcircled{2} P(a < x < 50) &= 0.1 \\ &= 1 - P(x < a) - P(x > 50) \end{aligned}$$

$$= 1 - P(x < a) - (1 - P(x < 50))$$

$$= P(x < 50) - P(x < a)$$

$$0.1 = 0.14713 - P(x < a)$$

$$P(x < a) = 0.04713$$

$$a = 46.4$$

$$\textcircled{3} \textcircled{1} i) X \sim B(1000, \frac{1}{6})$$

ii) n is large (1000)

$$\textcircled{6} X \sim N(167, 138.9)$$

$$P(a < X < 6) = 0.4$$



$$P(X > a) = 0.7 \rightarrow P(X < a) = 0.3$$

$$P(X < 6) = 0.7$$

$$a = 160$$

$$6 = 173$$

$$0.389$$